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Biogeographical Regions in the Aegean Archipelago: An Assessment for Freshwater Macroinvertebrates

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ABSTRACT

Aim: To delineate biogeographical regions in the Aegean Archipelago using freshwater macroinvertebrates with different dispersal abilities, and explore the relation of these regions to geological and environmental drivers.

Location: Aegean Archipelago, Greece.

Taxon: Freshwater macroinvertebrates: Odonata (damselflies/dragonflies), Trichoptera (caddisflies), and Plecoptera (stoneflies).

Methods: Species data information of three macroinvertebrate orders from the Aegean Islands was compiled to identify biogeographical regions. The clustering (UPGMA with Jaccard dissimilarity) method was applied to delineate regions. Differences among regions were tested using PERMANOVA on β -diversity turnover and nestedness. NMDS and Generalized Linear Models (GLMs) were used to assess the role of environmental variables in explaining the resulting biogeographical regions.

Results: Among islands, species turnover was the dominant component of β -diversity, indicating that compositional differences were primarily driven by species replacement. Six well-supported biogeographical regions reflecting both historical paleogeographic events and contemporary ecological patterns were identified. Differential dispersal abilities among groups also translated into biogeographical patterns, with Odonata showing weaker regional structuring than Trichoptera and Plecoptera. Western and eastern islands formed separate clusters, with central islands showing transitional patterns, reflecting past geological events in the area. Environmental gradients, particularly in precipitation, temperature, and freshwater availability, further shaped the biogeographical regions by influencing habitat stability and acting as ecological filters based on species' environmental tolerances.

Main Conclusions: This study provides the first freshwater biogeographical regionalization of the Aegean Archipelago using macroinvertebrate fauna. The six identified regions are ecologically and biogeographically meaningful, shaped by dispersal barriers, paleogeographic history, and local environmental variation. Turnover driven β -diversity and the contrast between high and low dispersal taxa highlight the value of trait-based approaches in biogeography.

1 | Introduction

Biogeographical regions are geospatial units with distinct faunistic and floristic assemblages. They facilitate the comprehension of the origin and the relation of biota they host (Kreft and Jetz 2010; Holt et al. 2013; Ficetola et al. 2017), and therefore are commonly used in biogeography, evolutionary ecology, and conservation studies (Kreft and Jetz 2010; Groves et al. 2002).

The delineation of biogeographical regions started many years ago (Humboldt and Bonpland 1807; Sclater 1858; Wallace 1876) and continues to pique the interest of many researchers, at both global (Vilhena and Antonelli 2015; Ficetola et al. 2017; He et al. 2017) and local scales (Kougiumoutzis et al. 2014, 2017; Oikonomou et al. 2014; Triantis et al. 2018). In 1876, Wallace divided the world into six major biogeographical regions based on terrestrial taxa (Sclater 1858).

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Later, Illies, in 1967 and 1978, introduced the first biogeographical European map specifically tailored for freshwater fauna, splitting Europe into 25 distinct regions. For Greece, this delineation included ecoregions six and seven: Hellenic western Balkan and Eastern Balkan, respectively (Illies 1967, 1978). The freshwater regions proposed by Illies still remain the principal standard reference for studies dealing with European freshwater fauna (Cervellini et al. 2020). However, it is noteworthy that these biogeographical regions have not escaped rigorous scrutiny and have faced significant criticisms (Abell et al. 2008; Zogaris et al. 2009).

Islands are ideal for the delineation of biogeographical regions because they are confined ecosystems with unique biodiversity composition due to key ecological and evolutionary processes (e.g., immigration, extinction, turnover, speciation) (MacArthur and Wilson 1967; Whittaker et al. 2017; Patiño et al. 2017). The Aegean Archipelago is one of the largest archipelagos in the world, comprising more than 7000 islands and islets (Triantis and Mylonas 2009), and is situated at the margins of the tectonic plates of Eurasia and Africa. Its complex paleogeographic history, combined with the high diversity and endemism, makes the Aegean Archipelago a natural laboratory for biogeographical studies (Trigas et al. 2013; Sfenthourakis and Triantis 2017). This has been a focus area for biogeographical research for long periods, but the majority of research has targeted terrestrial species of plants and animals. Botanists have made a substantial impact on the delineation of biogeographical regions in Greece (Strid 1996; Kougioumoutzis et al. 2017). Zoologists have also contributed to shaping the biogeographical boundaries in the Aegean Sea, based on terrestrial species (Triantis et al. 2018). Regarding aquatic animals, freshwater fishes are frequently used in biogeographical research (Oikonomou et al. 2014; Zogaris and Economou 2018), but no study has involved freshwater macroinvertebrates, even if they constitute 60% of the freshwater species (Balian et al. 2008).

Inland waters and freshwater biodiversity benefit society in multiple ways. They regulate local climate, natural hazards, and extreme events (flood protection, fire breaks, erosion prevention), air quality (carbon sequestration), and the quality of freshwater (water purification, breakdown, process, and transformation of pollutants, toxins, and heavy metals) (WWF 2022; Freshwater Information System 2024). The fact that almost 10% of global animal biodiversity is associated with habitats occupying <1% of the Earth's surface (Shiklomanov 1993) emphasises the threat posed by increasing human water demands (Dudgeon et al. 2006). Indeed, human activities have significant impacts on freshwater species and habitats, including many biodiversity hotspots (He et al. 2018). The threats to freshwater biodiversity include destruction or degradation of habitat, water pollution, river fragmentation, flow modification, overexploitation, and invasion by alien species (Dudgeon et al. 2006; Dudgeon 2019). Additionally, climate change further exacerbates the degradation of freshwater systems (Parmesan et al. 2022). Based on the Freshwater Living Planet Index (1970–2018), the average abundance of 1398 species worldwide declined by 83%, a rate twice that of biodiversity loss in terrestrial and marine realms (WWF 2022). Therefore, freshwater ecosystems may well be the most endangered ecosystems in the world, but despite the

large pressure, these systems still hold an incredible biodiversity. A better recognition of the benefits that humanity gains from freshwater biodiversity, combined with an effective and coordinated agenda for conservation and restoration, is essential for halting ongoing species declines and mitigating extinction risk (Lynch et al. 2023; Maasri et al. 2022; Darwall et al. 2018). Maasri et al. (2022) proposed an emergency recovery plan that includes restoring ecological connectivity, reducing pollution, and enhancing species protection as key strategies to reverse current trends.

Despite the ecological importance of freshwater biodiversity and the growing threat it faces, its biogeographical patterns remain understudied in insular regions such as the Aegean Archipelago. This study addresses this gap by (1) delineating biogeographical regions for freshwater macroinvertebrates across the archipelago using freshwater macroinvertebrates with different dispersal abilities, and (2) investigating abiotic drivers shaping these patterns. Dispersal ability plays a major role in bioregionalization (Christenhusz and Chase 2013). Taxa with high dispersal capacity often exhibit broader distributions and weaker spatial structuring, while poor dispersers tend to show stronger regional differentiation and higher turnover due to limited ability to overcome geographic barriers (Patrick et al. 2014; Heino 2013). To achieve our first objective, bioregionalization was obtained from three macroinvertebrate groups with varying dispersal abilities: Odonata, Trichoptera, and Plecoptera (Sarremejane et al. 2020). Odonata are considered good dispersers, with many adult species able to fly for even kilometres (up to 60 km; Wikelski et al. 2006). Trichoptera have a lower dispersal ability, but their aerial adults can still fly several hundred metres (up to 20 km; Salavert 2013), whereas Plecoptera are considered poor fliers with low dispersal abilities (up to 1 km; Briers et al. 2004). Accordingly, we expect taxa with limited dispersal capacity (e.g., Plecoptera and Trichoptera) should exhibit stronger bioregionalization, whereas more mobile taxa such as Odonata may show more homogeneous distributions due to their ability to overcome geographic isolation.

The paleogeographical evolution of the Aegean region, along with differences in current environmental characteristics among islands, is expected to drive the formation of distinct biogeographical regions. The region's complex paleogeographic history, which is shaped by tectonic movements, sea-level fluctuations, and long-term insularization, has resulted in a highly fragmented archipelago, with islands varying in size, altitude, geology, climatic conditions, and freshwater availability (Panitsa et al. 2018; Triantis et al. 2018). These environmental gradients, in combination with the spatial configuration of the archipelago (e.g., island clustering, distances from the mainland), are likely to influence both the composition and differentiation of freshwater communities. Accordingly, concerning our second objective, we expect geological and environmental drivers to be important in understanding the bioregionalization of freshwater macroinvertebrates in the Aegean region. This study marks a significant advancement in freshwater biogeography by providing the first regionalization of the Aegean Archipelago based on macroinvertebrate fauna. Findings of this study not only fill a major knowledge gap in island freshwater biogeography but also provide a valuable framework for biodiversity conservation in fragmented and climate-sensitive landscapes.

2 | Materials and Methods

2.1 | Species Distribution Data

Information on the distribution of freshwater macroinvertebrate species of Odonata, Trichoptera, and Plecoptera in the Aegean Islands was compiled. The data included species records obtained based on published sources enriched with our own data (Appendix S1). All records showing taxonomical uncertainties and doubtful presences were excluded. Also, as recommended by Triantis et al. (2018), only islands larger than 9 km² were considered to ensure adequate sampling completeness and ecological representativeness. These islands were excluded due to uneven sampling effort and their limited capacity to sustain permanent freshwater habitats.

2.2 | Environmental Variables

Four categories of environmental variables (current climate, topography, land use, and geology) were used to understand the delineation of biogeographical regions. These variables were selected because they represent key abiotic gradients known to influence biodiversity patterns and biogeographical regionalization in both terrestrial and freshwater systems (Panitsa et al. 2018; Cervellini et al. 2020; Soria et al. 2024). Current climatic predictors included mean annual precipitation (mm, AP; Figure S2.1) and mean annual temperature (°C, MAT; Figure S2.2), and were obtained from WorldClim v2 at a spatial resolution of 30 arc-seconds (~1 km; Fick and Hijmans 2017). Topographic predictors, including area (km, A), distance to mainland (km, DM), mean elevation (m, ME), and max elevation (m, MxE), were calculated in ArcGIS 10.1 (ESRI 2011, Redlands, CA, USA) using a Digital Elevation Model (DEM) with a spatial resolution of 5 × 5 m (Hellenic Cadastre 2020). Land use variables [artificial areas (ART.A.), agricultural areas (AGR.A.), pastures/grasslands (P.G.), forests (F), no vegetation (NO.VEG.), and water bodies (W.B)] were extracted from the CORINE Land Cover dataset (European Environment Agency 2018; CLC2018; 100 m spatial resolution; Figure S2.3), using ArcGIS 10.1 (ESRI 2011, Redlands, CA, USA). Finally, geology variables [recent sediments (S), carbonates (C), magmatic/metamorphic rocks (MMS), and volcanic rocks (V)] were derived from the Geological map of Greece (H.S.G.M.E. 2024; scale 1:50,000; Figure S2.4), also using ArcGIS 10.1 (ESRI 2011, Redlands, CA, USA). For each island, the area of each land use and geological category was calculated and converted to percentage cover.

2.3 | Data Analysis

To delineate regions, β -diversity was used. There are numerous metrics for calculating β -diversity (Anderson et al. 2011), and a considerable debate exists among scientists regarding their use in biogeographical studies (Leprieur and Oikonomou 2014). Based on presence-absence matrices (rows representing islands and columns representing species), the total β -diversity (β_{jac}) was computed. This index can be partitioned into two additive components: the turnover component (β_{itu}), which reflects species replacement, and the nestedness component (β_{jne}), which captures dissimilarity due to richness differences when one assemblage

is a subset of another (Baselga 2010). These β -diversity metrics were computed using the *beta.pair* and *beta.multi* functions of the 'betapart' package (Baselga and Orme 2012). Mean β -diversity values were calculated to compare the relative contribution of turnover and nestedness components among taxa and to guide the selection of β_{itu} for subsequent clustering analyses. Metrics that measure species replacement (β_{itu}) independently of differences in species richness have been suggested as more appropriate for bioregionalization (Kreft and Jetz 2010; Holt et al. 2013), especially for areas with large changes in richness (Mouillot et al. 2013). This diversity component is also well suited for bioregionalization analyses (Kreft and Jetz 2010; Baselga 2012; Leprieur and Oikonomou 2014) and especially for regions with high levels of endemism, like the Aegean Archipelago.

Hierarchical Clustering Analysis (HCA) using arithmetic averages (UPGMA) was performed on β_{itu} dissimilarity matrices for each taxonomic group and two combinations (all groups together and only Trichoptera and Plecoptera), using the *hclust* function (base stats package). HCA proves beneficial for biogeographical purposes due to the hierarchical arrangement of biogeographical regions (Escalante 2009). The UPGMA linkage method was selected to ensure comparability with previous freshwater biogeographical analyses in the Aegean region (Zogaris and Economou 2018; FEOU 2008), following earlier applications in global bioregionalization studies (Kreft and Jetz 2010). Although recent network-based approaches provide alternative ways to delineate biogeographical regions, UPGMA remains widely used for its interpretability and consistency with existing biogeographical studies (Denelle et al. 2025). The expected product is a dendrogram representing the relative distance between all islands based on their species composition. The cophenetic correlation coefficient was used to examine how well a given dendrogram maintains the pairwise distances found in the original distance matrix (Saraçlı et al. 2013). This coefficient ranges from 1 for perfectly correlated variables to -1 for anti-correlated variables, with uncorrelated variables having values near zero, and can be used to compare different cluster solutions (Sokal and Rohlf 1962). To identify the optimal number of clusters, the Kelley-Gardner-Sutcliffe (KGS) penalty function (*kgs* function, 'maptree' package; Kelley et al. 1996) was used. This method indicates the maximum differences among groups and maximum similarities within groups of islands based on the dissimilarity matrix. The minimum of KGS function corresponds to the optimal number of clusters. Out of the five dendrograms, the selected regionalization was obtained considering the cophenetic correlation coefficients along with the option that covered more islands (not all islands had representatives of all taxonomic groups). Silhouette index (Rousseeuw 1987) was used to evaluate the internal validity of the final hierarchical clustering solution obtained through the KGS criterion, using the *silhouette* function of the 'cluster' package (Maechler et al. 2025). To further assess the robustness of the delineated biogeographical regions, a network-based clustering analysis using the Louvain algorithm (*cluster_louvain* function, 'igraph' package; Csárdi et al. 2025) was applied to a β_{itu} similarity network derived from the same dissimilarity matrix used for the hierarchical analysis. The strength of the community structure within the network was quantified using modularity (Q; *modularity* function, 'igraph' package; Csárdi et al. 2025), whereas the congruence between the two methods was quantified using the Adjusted Rand

Index (*adjustedRandIndex* function, ‘mclust’ package; Scrucca et al. 2023) and the V-measure index (‘infotheo’ and ‘philen-tropy’ packages; Drost 2018; Meyer 2008).

PERMANOVA was conducted to explore significant differences in species composition among the predefined biogeographical regions. Analyses were performed in R using the *adonis* function of the ‘vegan’ package (Oksanen et al. 2015), based on 999 permutations and the β_{jtu} dissimilarity matrix.

A non-metric multidimensional scaling (NMDS) was applied using the *metaMDS* function of the ‘vegan’ package (Oksanen et al. 2015) to visualise compositional similarities among islands. NMDS provides an ordination of sites in a reduced-dimensional space where similar assemblages are plotted closer together, and dissimilar ones further apart, offering an independent way to assess the consistency of the HCA-derived regions. Stress values were used to evaluate the quality of the ordination, and the resulting NMDS configuration was overlaid with the HCA regions to visualise their relative distances. Additionally, environmental variables were plotted as vectors to explore potential relationships with the resulting regions.

To assess the role of environmental variables in explaining resulting biogeographical regions, the Variance Inflation Factor (VIF) was used to identify collinearity and exclude highly correlated variables, through a stepwise procedure, using the *vif* function of the ‘usdm’ package (Naimi et al. 2014). A threshold value of 5 was applied, following common practice in ecological modelling (Zuur et al. 2010; Dormann et al. 2013) and supported by recent methodological evaluations that regard $VIF=5$ as a practical indicator for potential multicollinearity (Marcoulides and Raykov 2019). Values of $VIF > 5$ suggest that the information contained in one variable is largely shared with others; however,

such values may still be acceptable when model coefficients remain statistically significant (Heiberger and Holland 2015). Finally, Generalized Linear Models (GLMs) with multinomial logit-link functions, using the *multinom* function of the ‘nnet’ package (Venables and Ripley 2002), and considering the resulting biogeographical regions as a multinomial response variable and the VIF-selected environmental variables (Rueda et al. 2010), were used to detect environmental drivers behind. The model selection was based on corrected Akaike's information criterion (AICc) scores (Johnson and Omland 2004), using the *dredge* function of the ‘MuMIn’ package (Barton 2023), with the selection criterion of $DAICc < 2$ (Burnham and Anderson 2002). To assess potential overfitting and evaluate model robustness, a leave—one out cross-validation (LOOCV) was performed using the *multinom* function. This cross—validation approach is considered suitable for small—sample ecological datasets (Fielding and Bell 1997).

All analyses were conducted in R (R 4.3.1 version, R Development Core Team 2023).

3 | Results

In total, 19,071 occurrence records were compiled, representing 283 species—72 Odonata, 159 Trichoptera, and 52 Plecoptera (see Table S2.1, S2.2 and Figure S2.5 for details). Odonata was present in 39 islands, Trichoptera in 22, and Plecoptera in 16.

The mean values of beta diversity components (β_{jtu} and β_{jne}) varied across the three taxonomic groups. Across all taxa, β_{jtu} consistently dominated, indicating that species turnover largely drives compositional dissimilarity, while β_{jne} played a smaller role, primarily in taxa with limited dispersal (Figure 1).

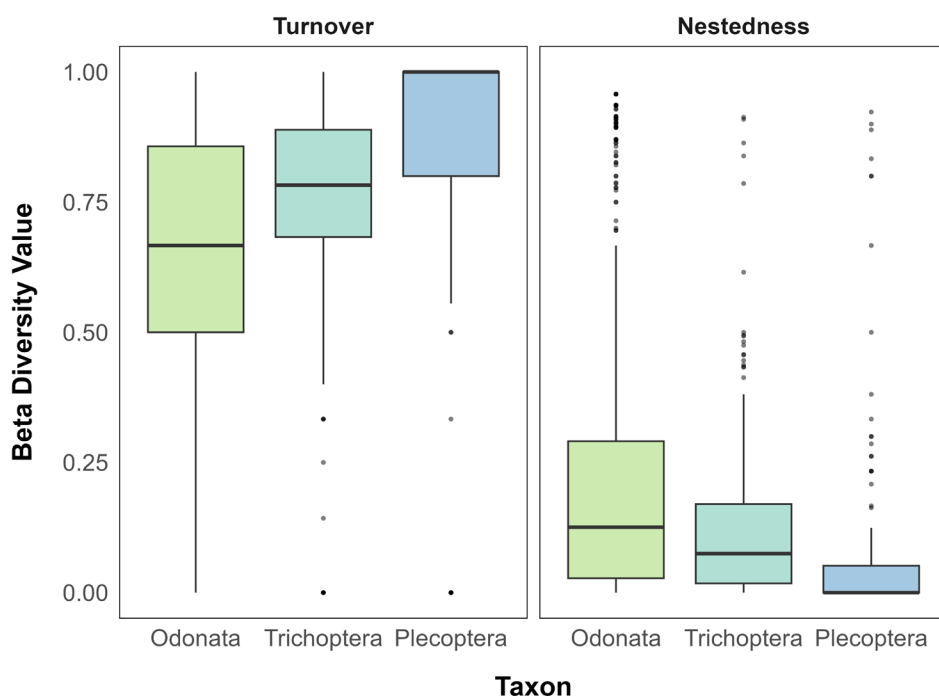


FIGURE 1 | Comparison of β -diversity components (species turnover: β_{jtu} and nestedness: β_{jne}) across all island pairs. Species turnover consistently dominated, justifying the emphasis on β_{jtu} in subsequent regional clustering.

3.1 | Biogeographical Regions

The UPGMA clustering analysis based on each taxonomic group separately showed differences both in the number of clusters and the islands per cluster (Odonata: Figure S2.6, Trichoptera: Figure S2.7, Plecoptera: Figure S2.8). Five biogeographical regions for Odonata were obtained, with a cophenetic correlation coefficient of 0.64 (Figure S2.6b) indicating limited evidence of bioregionalization. Instead, the coefficients were higher for Trichoptera (0.78; Figure S2.7b), Plecoptera (0.84; Figure S2.8b), and both together (0.81; Figure 2; Figure S2.9), but lower for the three groups together (0.73; Figure S2.10b). A

summary of the clustering results and cophenetic correlation coefficients for each taxonomic group is provided in Table S2.3. The high coefficient for Trichoptera and Plecoptera, along with a higher number of islands considered in this option, resulted in a selected regionalization of six biogeographical regions: Northern Sporades, Skyros, Central Aegean, Crete and Southeastern Aegean, Northeastern Aegean, and Thrace (Figure 2a,b). The big separation between islands in eastern and western Aegean emerged in the dendrogram at the height of coefficient 0.85 (Figure 2a, node 5). The amalgamation of the islands of Central Aegean with Crete and Karpathos occurred at coefficient of 0.74 (Figure 2a, node 1). Also, the islands of

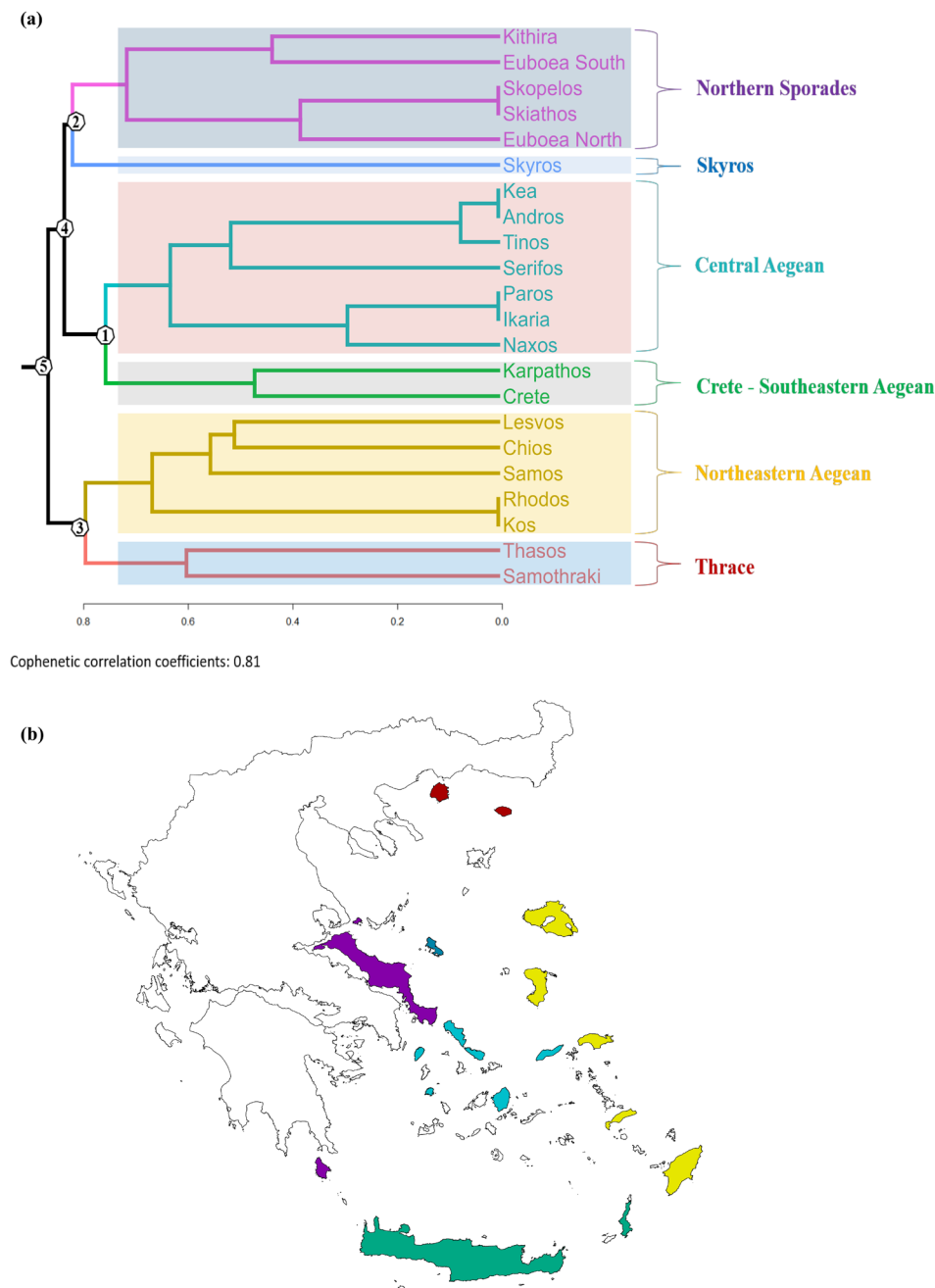


FIGURE 2 | Dendrogram using UPGMA (a) and map of the Aegean Archipelago with the biogeographical regions identified for Trichoptera and Plecoptera (b). Colours of the islands correspond to those in the dendrogram: purple for Northern Sporades, blue for Skyros, cyan for Central Aegean, green for Crete and Southeastern Aegean, yellow for Northeastern Aegean, and red for Thrace.

Northeastern Aegean and Thracian Sea were merged last at coefficient of 0.81 (Figure 2a, node 3).

The regionalization was validated using the Silhouette index, which yielded a mean silhouette width of 0.27 (Figure S2.11), indicating a moderate level of internal consistency among biogeographical regions. In addition, the Louvain clustering produced three biogeographical regions with a modularity value of $Q=0.441$ (Figure S2.12), indicating a well-structured community organisation. Comparison between the hierarchical and network-based methods showed an Adjusted Rand Index of 0.657 (Figure S2.13), reflecting a substantial level of correspondence between the two clustering approaches. The spatial association between the two regionalization schemes (V-measure index) was 0.76, further supporting the high level of spatial congruence between them. The three Louvain modules correspond to broader biogeographical entities that, in the hierarchical clustering, are further subdivided into two subregions each, resulting in the six biogeographical regions identified by the hierarchical analysis. Significant PERMANOVA results among the six regions were found for β jt for all three taxonomic groups. Differences were significant but lower for Odonata ($F=1.70$, $p=0.035$) than for Trichoptera ($F=4.71$, $p=0.001$) or Plecoptera ($F=4.13$, $p=0.001$), indicating taxon-specific patterns of regional differentiation.

3.2 | Drivers of Biogeographical Regions

The six selected regions were also separated in the NMDS (Figure 3), with a stress value of 0.20. Mean annual precipitation

(AP; $p=0.001$) separated the islands in the Northeastern Aegean region (indicated with yellow in the plot) from the islands in the very dry environment of the Central Aegean region (Cyclades, with the blue colour in the plot). Also, the distance to the mainland (DM; $p=0.005$) separated the “continental” islands (islands which separated more recently from the mainland, Western Aegean, Northern Sporades, and Thrace) from the other islands.

Based on VIF scores, six environmental predictors were used for the multiple regression GLMs (Table 1, Table S2.4). AP explained the highest (30%) amount of deviance, followed by distance to

TABLE 1 | Results of the best model fit exploring the influence of the predictors on biogeographical regions in Aegean Islands.

Variables	Best model	R^2
Distance to mainland (DM)	+	0.29
Maximum elevation (MxE)	–	0.15
Mean annual temperature (MAT)	+	0.22
Mean annual precipitation (AP)	+	0.30
Magmatic/metamorphic rocks (MMS)	–	0.26
Carbonates (C)	+	0.09
AIC	50.00	–
R^2 model	0.99	–

Note: The symbol ‘+’ indicates variables included in the best model ($\Delta AIC_c < 2$), the R^2 is the deviance explained by each predictor, and the R^2 model is the deviance explained by the best model.

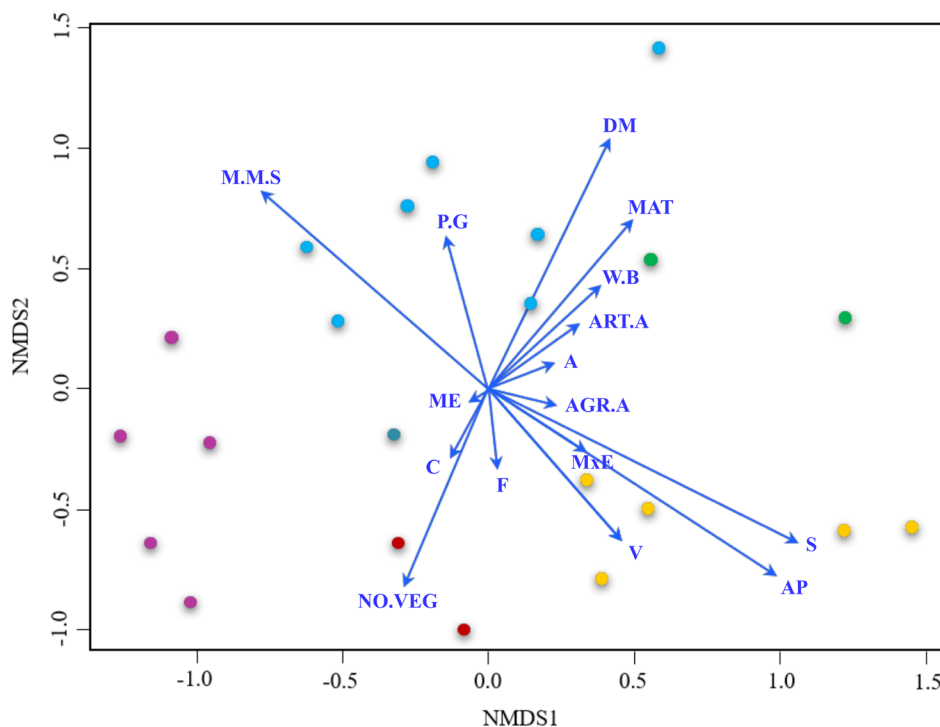


FIGURE 3 | NMDS ordination based on the β jt dissimilarity matrix. The stress value is 0.20. Each dot corresponds to an island and colours to the six biogeographical regions identified in the dendrogram of Figure 2: purple for Northern Sporades, blue for Skyros, cyan for Central Aegean, green for Crete and Southeastern Aegean, yellow for Northeastern Aegean, and red for Thrace. Blue arrows represent significant environmental variables ($p < 0.05$), including climatic (AP, MAT), topographic (DM, A, ME, MxE), land use (e.g., ART.A, AGR.A, NO.VEG), and geological factors (e.g., S, C, M.M.S., V). Arrow direction and length indicate correlation strength with NMDS axes.

mainland (DM), magmatic/metamorphic rocks (MMS), mean annual temperature (MAT), and carbonates (C) (Table 1). The best-fitted model included four expected indicators (AP, MAT, DM, and C), explaining 99% of the deviance. However, cross-validation analysis yielded a classification accuracy of 68.2%, suggesting that while the model fit the data very well, its predictive capacity was moderate—indicating no overfitting but rather an expected limitation due to the small sample size and class imbalance among regions.

4 | Discussion

Bioregionalization of the Aegean Archipelago based on species data of freshwater macroinvertebrates revealed six biologically relevant regions: Northern Sporades, Skyros, Central Aegean, Crete and Southeastern Aegean, Northeastern Aegean, and Thrace. To our knowledge, this is the first study about the delineation of freshwater biogeographical regions of the Aegean Archipelago considering macroinvertebrate fauna. The identified regions align well with some previous biogeographical studies based on freshwater fishes in Greece (Zogaris and Economou 2018; FEOW 2008). Nevertheless, certain inconsistencies became apparent. Specifically, the Central Aegean, Skyros, and the Northern Sporades regions are grouped into a single unit in their classification, whereas our results suggest that they represent distinct biogeographical regions. Additionally, in their framework, Kithira is grouped with the mainland, while in our analysis, it is associated with the Northern Sporades. Notably, the analysis by Zogaris and Economou (2018) was based solely on data from the mainland, without incorporating raw data from the Aegean islands. For the islands, they relied on expert judgement using paleogeographic reconstructions of the Aegean Archipelago, which may account for some discrepancies with our findings.

In agreement with our first expectation, the differences observed in regionalization among the taxonomic groups aligned closely with their dispersal abilities. Regionalization was more consistent for Trichoptera and Plecoptera than for Odonata. The high dispersal ability of Odonata likely facilitates more continuous movement among islands, overcoming geographic barriers and maintaining wider distributions, which is also in agreement with the observed lower turnover and higher nestedness. In contrast, the limited dispersal ability of Plecoptera likely enhances community differentiation by increasing isolation among islands, making local environmental conditions more influential in shaping species composition. Trichoptera, with moderate dispersal capacity, showed intermediate levels of turnover, which may reflect a balance between some connectivity and strong environmental filtering. This emphasises that the proposed regions are particularly shaped by taxa constrained by dispersal barriers, highlighting the value of trait-based approaches in island biogeography (Tickner et al. 2020). These results also indicate that species replacement between islands is the primary process shaping compositional dissimilarity, which is consistent with expectations for island systems where isolation and limited connectivity lead to distinct local communities (Whittaker et al. 2008; Soria et al. 2024).

Our findings further align with our second expectation, indicating that the resulting biogeographical regions represent

geologically and ecologically meaningful units. This is strongly supported by the congruence between the derived regions and the well-known paleogeographic history of the Aegean (Hausdorf and Hennig 2005). Key geological events, such as the formation of the Mid-Aegean Trench (~12 Ma; Dermitzakis and Papanikolaou 1981), the Messinian Salinity Crisis (~6–5.3 Ma; Duggen et al. 2003) and the post-glacial sea level rise during the Late Pleistocene to Early Holocene (Kapsimalis et al. 2009), played a central role in shaping the insular landscape and the degree of isolation among island groups. These long-term processes led to both ancient and recent insularization patterns, reflected in the varying levels of faunal distinctiveness among regions (Triantis et al. 2018). Regarding ecological factors explaining the resulting biogeographical regions, the environmental variables considered aimed to identify broad-scale abiotic gradients. Our approach effectively captured the main environmental dimensions influencing biodiversity across the Aegean islands, but incorporating additional variables (e.g., CHELSA, Karger et al. 2017; MEDIS, Santi et al. 2024) could reveal unseen patterns and increase the robustness of our analysis.

The biogeographical region referred to here as ‘Northern Sporades’, as defined by our clustering analysis, includes not only the Northern Sporades islands *sensu stricto*, but also other nearby islands such as Euboea and Kithira. Its delineation aligns with previous ichthyogeographical studies (Zogaris and Economou 2018) and phytogeographical regionalizations (Kougiumoutzis et al. 2017), which also highlighted affinities between Western Aegean islands and adjacent mainland regions. Geologically, this region is characterised by recent insularization, as most of the islands became separated from the mainland only during the late Pleistocene to early Holocene, often as a result of post-glacial sea level rise (Lambeck 1995). This recent fragmentation explains the high ecological similarity among the islands and with nearby continental watersheds. Despite the geographical spread of this region, its internal coherence is supported by shared climatic and geological features, such as limestone-dominated landscapes, seasonally dry hydrology and the presence of limited but perennial freshwater habitats (Sfenthourakis and Triantis 2017). The distinctiveness of this biogeographical region is further reinforced by the topographical relief and hydrological isolation of the individual drainage basins. Overall, the region exhibits a well-defined ecological identity, shaped by both historical continuity and recent patterns of isolation (Iliadou et al. 2014; Panitsa et al. 2010).

Our analysis revealed Skyros as a biogeographically distinct region. When analysed separately, Trichoptera assemblages on Skyros clustered with Euboea North, while Plecoptera assemblages aligned more closely with Kithira and Euboea South (Appendix S2: Figure S3). This taxon-specific grouping suggests differential dispersal and colonisation histories among freshwater macroinvertebrates, a pattern observed in other studies of Aegean freshwater organisms (Hupało et al. 2020). Skyros's unique position in the Aegean, at the intersection of four other biogeographical regions (Northern Sporades, Northeastern Aegean, Thrace, and Central Aegean), likely contributes to its distinctive faunal composition. Despite its relatively low species richness (seven species, one of which is endemic), Skyros shares taxa with multiple neighbouring regions. According to Dermitzakis (1990), Skyros is located on a separate continental

shelf southeast of the main Northern Sporades chain and has likely been isolated for a longer period. This paleogeographical separation has also been noted in studies on lizards, where Skyros and its nearby islets have traditionally been considered separately due to their distinct evolutionary and biogeographical histories (Triantis et al. 2005; Pafilis et al. 2013; Fofopoulous et al. 2024). Furthermore, these findings are supported by broader biogeographical analyses using other taxonomic groups. For instance, Kougiumoutzis et al. (2017) identified Skyros as a distinct or transitional unit between Northern Sporades and Central Aegean. Thus, while Skyros may function as an ecological bridge among surrounding regions, it also exhibits enough distinctiveness to be recognised as an independent biogeographical unit within the Aegean Archipelago. Topographically and climatically, Skyros exhibits significant heterogeneity. Its moist, vegetated northern sector closely resembles the Northern Sporades, while its drier, less vegetated southern side exhibits affinities with the Central Aegean (Vaitis et al. 2017, 2019).

The Central Aegean islands, primarily the Cyclades, formed a distinct and cohesive biogeographical region in our analysis. This pattern is consistent with their shared paleogeographic history. During the Late Pleistocene, particularly at the Last Glacial Maximum (ca. 18ka), a large landmass known as the Cyclades Plateau was exposed due to sea level regressions. This plateau, encompassing much of today's Central Aegean, remained largely connected until the end of the Pleistocene (ca. 14–9ka), when rising sea level progressively fragmented it into the present-day island configuration (Kapsimalis et al. 2009; Van Andel and Shackleton 1982; Lambeck 1995). This long period of geological unity explains the strong internal similarity in freshwater macroinvertebrate assemblages across the Central Aegean. Most dispersal likely occurred before the complete isolation of the islands, resulting in persistent biotic homogeneity. The grouping of Ikaria, typically considered part of the Northeastern Aegean, with the Central Aegean islands in our analysis may reflect either ecological similarity or instances of dispersal across the Mid-Aegean Trench, as has been documented in various taxa, including isopods, land snails, beetles, and reptiles (Poulakakis et al. 2015). Some paleogeographic reconstructions (e.g., Creutzburg 1963) also suggest the existence of ephemeral land bridges across the Mid-Aegean Trench, which may have facilitated such biotic exchanges. In this context, the Central Aegean's biogeographical coherence arises from both its paleogeographic continuity and the relatively recent onset of insularity, which together shaped the shared biodiversity patterns observed today. The delineation of the Central Aegean as a distinct biogeographical region is also supported by other recent studies based on different taxonomic groups, such as vascular plants (Kougiumoutzis et al. 2017).

Crete and the island in the Southwestern Aegean, Karpathos, form another distinct and well-supported biogeographical region. This grouping reflects both the long-term geological isolation of these islands and their specific ecological conditions, which set them apart from other parts of the Aegean Archipelago. Crete has long been recognised as one of the most biogeographically unique islands in the Mediterranean, with high levels of endemism and taxonomic originality across multiple taxa (Sfenthourakis and Triantis 2017; Triantis et al. 2018). Its separation from the Peloponnese occurred approximately 5

million years ago, during the Messinian Salinity Crisis (Armijo et al. 1992), while Karpathos became isolated from Rhodos and Anatolia shortly thereafter (Daams and Van der Weerd 1980). These long-standing insular conditions, combined with complex topography and varied freshwater microhabitats (e.g., springs, karstic systems), have fostered the development of distinct freshwater faunas. Climatically, this region lies within the warmest and driest parts of the Aegean, with low annual precipitation and long dry seasons, particularly in Crete and southern Karpathos. These conditions result in highly fragmented and seasonal freshwater systems, favouring species that are either resilient to drying or adapted to isolated spring habitats. This hydrological and climatic context likely contributes to the observed distinctiveness of macroinvertebrate assemblages. The biogeographical uniqueness of this region is supported not only by our data, but also by studies on other taxa. For instance, Kougiumoutzis et al. (2017), Triantis et al. 2018 and Zogaris and Economou (2018) highlighted the distinctiveness of Crete and parts of the Southeastern Aegean in both botanical and zoological datasets. Crete, in particular, has been treated as a standalone biogeographical unit in several continental and insular biogeography frameworks due to its high number of endemics and evolutionary relicts. Taken together, the isolation, climatic constraints, and ecological filtering in this region have likely driven the formation of a distinct biogeographical identity.

The Northeastern Aegean region includes islands such as Lesvos, Chios, Samos, Rhodos and Kos, which have a much older and persistent insularity than other Aegean islands. The region's unique biogeographical identity is primarily shaped by two major geological events: the Mid-Aegean Trench and the Messinian Salinity Crisis. Following the reflooding of the Mediterranean Basin, the northeastern Aegean islands became permanently insular (~5.3 Ma), maintaining prolonged isolation that likely underpins their distinct freshwater macroinvertebrates. Nevertheless, some of these islands remained connected to the Asia Minor mainland until relatively recently (~20ka BP; Perissoratis and Conispoliatis 2003), allowing for intermittent faunal exchanges before their final isolation. Importantly, the Northeastern Aegean islands receive higher annual precipitation and have lower mean temperatures than the more arid and warmer Central and Southeastern Aegean (Vaitis et al. 2019). These climatic gradients contribute to the development and persistence of distinct freshwater habitats and influence macroinvertebrate diversity. Our findings are consistent with a wide range of biogeographical studies based on other taxonomic groups, including reptiles (Lymberakis et al. 2018), tenebrionid beetles (Fattorini 2002), isopods (Sfenthourakis 1996), scorpions (Parmakelis et al. 2022), land snails (Parmakelis and Mylonas 2004), vascular plants (Kougiumoutzis et al. 2017), and freshwater fishes (Zogaris and Economou 2018). A multi-taxa approach (Triantis et al. 2018) has also corroborated the distinctiveness of this region.

Finally, the clustering of Thasos and Samothraki into a distinct biogeographical region (Thrace) represents a more continental biogeographical pattern. Despite their insular status today, both islands are geologically young and were connected to the mainland until relatively recently in geological terms, particularly during Late Pleistocene sea-level lowstands (Lykousis 2009).

This recent separation likely explains the strong faunal similarity between these islands and the nearby mainland freshwater systems (Hupało et al. 2020). Their proximity to the mainland likely facilitates occasional natural dispersal, either through passive mechanisms or via land connections during glacial periods. Climatically, this region differs markedly from the more arid Central and Southern Aegean. It experiences higher annual rainfall, lower mean temperatures, and the presence of relatively stable freshwater habitats, particularly on Samothraki, which has permanent streams. These features may promote the persistence of species with narrow ecological tolerances and help sustain freshwater taxa not found further south. From a biogeographical perspective, the inclusion of these islands as a unique freshwater region is consistent with other studies across multiple taxa. Zogaris and Economou (2018), in their fish-based regionalization, also placed Samothraki and the Evros basin in a Thracian unit. Similarly, Kougioumoutzis et al. (2017) identified this northeastern sector as a distinct phytogeographic cluster within the Aegean. The convergence of results across unrelated taxonomic groups reinforces the robustness of this regional delineation.

Our study provides a novel and integrative approach to freshwater biogeographical regionalization in island systems, emphasising the importance of beta diversity and dispersal traits in shaping spatial community patterns. The findings highlight the interplay between historical paleogeographic events and contemporary ecological processes in driving biogeographical differentiation. This framework can be applied to other regions and taxonomic groups, offering a robust foundation for biodiversity conservation and management strategies in fragmented and diverse freshwater ecosystems.

Author Contributions

Anastasia Lampou: conceptualisation, methodology, data collection/investigation, sampling, taxonomy, formal analysis, writing – original draft. **Núria Bonada:** conceptualisation, methodology, writing – review and editing, supervision, funding acquisition. **Nikolaos Skoulidakis:** writing – review and editing, supervision, funding acquisition. All authors reviewed and edited the final version.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study is available in the [Supporting Information](#) of this article.

References

- Abell, R., M. L. Thieme, C. Revenga, et al. 2008. “Freshwater Ecoregions of the World: A New Map of Biogeographic Units for Freshwater Biodiversity Conservation.” *Bioscience* 58, no. 5: 403–414. <https://doi.org/10.1641/B580507>.
- Anderson, M. J., T. O. Crist, J. M. Chase, et al. 2011. “Navigating the Multiple Meanings of β Diversity: A Roadmap for the Practicing Ecologist.” *Ecology Letters* 14: 19–28. <https://doi.org/10.1111/j.1461-0248.2010.01552.x>.
- Armijo, R., H. Lyon-Caen, and D. Papanastassiou. 1992. “E-W Extension and Holocene Normal Fault Scarps in the Hellenic Arc.” *Geology* 20: 491–494. [https://doi.org/10.1130/0091-7613\(1992\)020<0491:EWEAHN>2.3.CO;2](https://doi.org/10.1130/0091-7613(1992)020<0491:EWEAHN>2.3.CO;2).
- Balian, E. V., C. Lévêque, H. Segers, and K. Martens. 2008. “The Freshwater Animal Diversity Assessment: An Overview of the Results.” *Hydrobiologia* 595: 627–637. <https://doi.org/10.1007/s10750-007-9246-3>.
- Barton, K. 2023. “MuMIn: Multi-Model Inference. R Package Version 1.47.1.” <https://CRAN.R-project.org/package=MuMIn>.
- Baselga, A. 2010. “Partitioning the Turnover and Nestedness Components of Beta Diversity.” *Global Ecology and Biogeography* 19, no. 1: 134–143. <https://doi.org/10.1111/j.1466-8238.2009.00490.x>.
- Baselga, A. 2012. “The Relationship Between Species Replacement, Dissimilarity Derived From Nestedness, and Nestedness.” *Global Ecology and Biogeography* 21: 1223–1232. <https://doi.org/10.1111/j.1466-8238.2011.00756.x>.
- Baselga, A., and C. D. L. Orme. 2012. “Betapart: An R Package for the Study of Beta Diversity.” *Methods in Ecology and Evolution* 3: 808–812. <https://doi.org/10.1111/j.2041-210X.2012.00224.x>.
- Briers, R. A., J. H. R. Gee, H. M. Cariss, and R. Geoghegan. 2004. “Inter-Population Dispersal by Adult Stoneflies Detected by Stable Isotope Enrichment.” *Freshwater Biology* 49: 425–431. <https://doi.org/10.1111/j.1365-2427.2004.01198.x>.
- Burnham, K. P., and D. R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. Springer. <https://doi.org/10.1007/b97636>.
- Cervellini, M., P. Zannini, M. Di Musciano, et al. 2020. “A Grid-Based Map for the Biogeographical Regions of Europe.” *Biodiversity Data Journal* 8: e53720. <https://doi.org/10.3897/BDJ.8.e53720>.
- Christenhusz, M. J. M., and M. W. Chase. 2013. “Biogeographical Patterns of Plants in the Neotropics – Dispersal Rather Than Plate Tectonics Is Most Explanatory.” *Botanical Journal of the Linnean Society* 171, no. 1: 277–286. <https://doi.org/10.1111/j.1095-8339.2012.01301.x>.
- Creutzburg, N. 1963. “Die Palliogeographische Entwicklung der Insel Kreta vom Miozan bis zur Gegenwart.” *Kritika Chronika* 15-16: 336–344.
- Csárdi, G., T. Nepusz, V. Traag, et al. 2025. “igraph: Network Analysis and Visualization in R.” <https://doi.org/10.5281/zenodo.7682609>.
- Daams, R., and A. V. Van derWeerd. 1980. “Early Pliocene Small Mammals From the Aegean Island of Karpathos (Greece) and Their Paleogeographic Significance.” *Geologie en Mijnbouw* 59: 27–331.
- Darwall, W., V. Bremerich, A. De Wever, et al. 2018. “The Alliance for Freshwater Life: A Global Call to Unite Efforts for Freshwater

- Biodiversity Science and Conservation." *Aquatic Conservation: Marine and Freshwater Ecosystems* 28, no. 4: 1015–1022. <https://doi.org/10.1002/aqc.2958>.
- Denelle, P., B. Leroy, and M. Lenormand. 2025. "Bioregionalization Analyses With the Bioregion R Package." *Methods in Ecology and Evolution* 16, no. 3: 496–506. <https://doi.org/10.1111/2041-210X.14496>.
- Dermitzakis, D. M. 1990. "Paleogeography, Geodynamic Processes and Event Stratigraphy During the Late Cenozoic of the Aegean Area." *Academia Nazionale Lincei* 85: 263–288.
- Dermitzakis, D. M., and D. J. Papanikolaou. 1981. "Paleogeography and Geodynamics of the Aegean Region During the Neogene." *Annales Geologique Des Pays Helleni* 30: 245–289.
- Dormann, C. F., J. Elith, S. Bacher, et al. 2013. "Collinearity: A Review of Methods to Deal With It and a Simulation Study Evaluating Their Performance." *Ecography* 36: 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>.
- Drost, H. G. 2018. "philentropy: Similarity and Distance Quantification Between Probability Functions. R Package Version 0.6.1." <https://CRAN.R-project.org/package=philentropy>.
- Dudgeon, D. 2019. "Multiple Threats Imperil Freshwater Biodiversity in the Anthropocene." *Current Biology* 29, no. 19: 960–967. <https://doi.org/10.1016/j.cub.2019.08.002>.
- Dudgeon, D., A. H. Arthington, M. O. Gessner, et al. 2006. "Freshwater Biodiversity: Importance, Threats, Status and Conservation Challenges." *Biological Reviews* 81: 163–182. <https://doi.org/10.1017/S1464793105006950>.
- Duggen, S., K. Hoernle, P. den van Bogaard, L. Rupke, and J. P. Morgan. 2003. "Deep Roots of the Messinian Salinity Crisis." *Nature* 422: 602–606. <https://doi.org/10.1038/nature01553>.
- Environmental Systems Research Institute (ESRI). 2011. *ArcGIS Desktop: Release 10, Redlands*. Environmental Systems Research Institute.
- Escalante, T. 2009. "Un Ensayo Sobre Regionalización Biogeográfica." *Revista Mexicana de Biodiversidad* 80: 551–560.
- European Environment Agency. 2018. "Copernicus Land Monitoring Service CLC2018, V20b2." <https://doi.org/10.2909/960998c1-1870-4e82-8051-6485205ebbacc>.
- Fattorini, S. 2002. "Biogeography of the Tenebrionid Beetles (Coleoptera, Tenebrionidae) on the Aegean Islands (Greece)." *Journal of Biogeography* 29: 49–67. <https://doi.org/10.1046/j.1365-2699.2002.00656.x>.
- FEOW. 2008. "Freshwater Ecoregions of the World-Conservation Science Program." The Nature Conservancy and World Wildlife Fund, Washington, DC. <https://www.feow.org/>.
- Ficetola, G., F. Mazel, and W. Thuiller. 2017. "Global Determinants of Zoogeographical Boundaries." *Nature Ecology & Evolution* 1: 0089. <https://doi.org/10.1038/s41559-017-0089>.
- Fick, S. E., and R. J. Hijmans. 2017. "WorldClim 2: New 1 km Spatial Resolution Climate Surfaces for Global Land Areas." *International Journal of Climatology* 37, no. 12: 4302–4315. <https://doi.org/10.1002/joc.5086>.
- Fielding, A. H., and J. F. Bell. 1997. "A Review of Methods for the Assessment of Prediction Errors in Conservation Presence/Absence Models." *Environmental Conservation* 24, no. 1: 38–49. <https://doi.org/10.1017/S0376892997000088>.
- Foufopoulos, J., S. Roussos, S. Kalogiannis, S. Kalb, I. Strachinis, and K. M. Brock. 2024. "The Herpetofauna of the Sporades Islands (Aegean Sea, Greece): New Discoveries and a Review of a Century of Research." *Herpetozoa* 37: 241–258. <https://doi.org/10.3897/herpetozoa.37.e125965>.
- Freshwater Information System [FIS]. 2024. "Regulating and Maintaining Ecosystem Services." <http://fis.freshwatertools.eu/index.php/infolib/ecoservices/regulating-services.html>.
- Groves, C. R., D. B. Jensen, L. L. Valutis, et al. 2002. "Planning for Biodiversity Conservation: Putting Conservation Science Into Practice: A Seven-Step Framework for Developing Regional Plans to Conserve Biological Diversity, Based Upon Principles of Conservation Biology and Ecology, Is Being Used Extensively by the Nature Conservancy to Identify Priority Areas for Conservation." *Bioscience* 52: 499–512. [https://doi.org/10.1641/0006-3568\(2002\)052\[0499:PFBCPC\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2002)052[0499:PFBCPC]2.0.CO;2).
- Hausdorf, B., and C. Hennig. 2005. "The Influence of Recent Geography, Palaeogeography and Climate on the Composition of the Fauna of the Central Aegean Islands." *Biological Journal of the Linnean Society* 84: 785–795. <https://doi.org/10.1111/j.1095-8312.2005.00467.x>.
- He, F., V. Bremerich, C. Zarfl, et al. 2018. "Freshwater Megafauna Diversity: Patterns, Status and Threats." *Diversity and Distributions* 24, no. 10: 1395–1404. <https://doi.org/10.1111/ddi.12780>.
- He, J., H. Kreft, E. Gao, Z. Wang, and H. Jiang. 2017. "Patterns and Drivers of Zoogeographical Regions of Terrestrial Vertebrates in China." *Journal of Biogeography* 44: 1172–1184. <https://doi.org/10.1111/jbi.12892>.
- Heiberger, R. M., and B. Holland. 2015. *Statistical Analysis and Data Display: An Intermediate Course With Examples in R*. 2nd ed. Springer-Verlag. <https://doi.org/10.1007/978-1-4939-2122-5>.
- Heino, J. 2013. "The Importance of Metacommunity Ecology for Environmental Assessment Research in the Freshwater Realm." *Biological Reviews* 88: 166–178. <https://doi.org/10.1111/j.1469-185X.2012.00244.x>.
- Hellenic Cadastre S.A. 2020. "Digital Elevation Model-DEM-LSO (5m)." <https://www.hkmg.gov.gr/>.
- Hellenic Survey of Geology and Mineral Exploration (H.S.G.M.E.). 2024. "Geological Map of Greece, Scale 1:50.000. Metadata ID: 251fca3a-1b32-11ef-a4ef-00d861c828db." <https://catalog.eagme.gr/geonetwork/srv/eng/catalog.search#/metadata/251fca3a-1b32-11ef-a4ef-00d861c828db>.
- Holt, B. G., J. P. Lessard, M. K. Borregaard, et al. 2013. "An Update of Wallace's Zoogeographic Regions of the World." *Science* 339: 74–78. <https://doi.org/10.1126/science.1228282>.
- Humboldt, A., and A. Bonpland. 1807. *Essay on the Geography of Plants*. University of Chicago Press.
- Hupało, K., I. Karaouzas, T. Mamos, and M. Grabowski. 2020. "Molecular Data Suggest Multiple Origins and Diversification Times of Freshwater Gammarids on the Aegean Archipelago." *Scientific Reports* 10: 19813. <https://doi.org/10.1038/s41598-020-75802-2>.
- Iliadou, E., A. Kallimanis, P. Dimopoulos, and M. Panitsa. 2014. "Comparing the Two Greek Archipelagos Plant Species Diversity and Endemism Patterns Highlight the Importance of Isolation and Precipitation as Biodiversity Drivers." *Journal of Biological Research* 21: 16–25. <https://doi.org/10.1186/2241-5793-21-16>.
- Illies, J. 1967. *Limnofauna Europaea. Eine Zusammenstellung Aller Die Europäischen Binnengewässer Bewohnenden Mehrzelligen Tierarten Mit Angaben Über Ihre Verbreitung und Ökologie*. Gustav Fischer Verlag.
- Illies, J. 1978. *Limnofauna Europaea*. Gustav Fisher Verlag.
- Johnson, J. B., and K. S. Omland. 2004. "Model Selection in Ecology and Evolution." *Trends in Ecology & Evolution* 19: 101–108. <https://doi.org/10.1016/j.tree.2003.10.013>.
- Kapsimalis, V., K. Pavlopoulos, I. Panagiotopoulos, et al. 2009. "Geoarchaeological Challenges in the Cyclades Continental Shelf (Aegean Sea)." *Zeitschrift für Geomorphologie* 53, no. 1: 169–190. <https://doi.org/10.1127/0372-8854/2009/0053S1-0169>.
- Karger, D. N., O. Conrad, J. Böhner, et al. 2017. "Climatologies at High Resolution for the Earth's Land Surface Areas." *Scientific Data* 4, no. 1: 1–20. <https://doi.org/10.1038/sdata.2017.122>.
- Kelley, L. A., S. P. Gardner, and M. J. Sutcliffe. 1996. "An Automated Approach for Clustering an Ensemble of NMR-Derived Protein Structures Into Conformationally Related Subfamilies." *Protein Engineering* 9: 1063–1065. <https://doi.org/10.1093/protein/9.11.1063>.

- Kougioumoutzis, K., S. M. Simaiakis, and A. Tiniakou. 2014. "Network Biogeographical Analysis of the Central Aegean Archipelago." *Journal of Biogeography* 41: 1848–1858. <https://doi.org/10.1111/jbi.12342>.
- Kougioumoutzis, K., A. T. Valli, E. Georgopoulou, S. M. Simaiakis, K. A. Triantis, and P. Trigas. 2017. "Network Biogeography of a Complex Island System: The Aegean Archipelago." *Journal of Biogeography* 44: 651–660. <https://doi.org/10.1111/jbi.12920>.
- Kreft, H., and W. Jetz. 2010. "A Framework for Delineating Biogeographical Regions Based on Species Distributions." *Journal of Biogeography* 37: 2029–2053. <https://doi.org/10.1111/j.1365-2699.2010.02375.x>.
- Lambeck, K. 1995. "Late Pleistocene and Holocene Sea-Level Change in Greece and South-Western Turkey: Separation of Eustatic, Isostatic, and Tectonic Contributions." *Geophysical Journal International* 122, no. 3: 1022–1044. <https://doi.org/10.1111/j.1365-246X.1995.tb06853.x>.
- Leprieur, F., and A. Oikonomou. 2014. "The Need for Richness Independent Measures of Turnover When Delineating Biogeographical Regions." *Journal of Biogeography* 41: 417–420. <https://doi.org/10.1111/jbi.12266>.
- Lykousis, V. 2009. "Sea-Level Changes and Shelf Break Prograding Sequences During the Last 400 ka in the Aegean Margins: Subsidence Rates and Palaeogeographic Implications." *Continental Shelf Research* 29: 2037–2044. <https://doi.org/10.1016/j.csr.2008.11.005>.
- Lymberakis, P., P. Pafilis, N. Poulakakis, K. Sotiropoulos, and E. Valakos. 2018. "Amphibians and Reptiles of the Aegean Sea." In *Biogeography and Biodiversity of the Aegean*. In Honour of Prof. Moysis Mylonas, edited by S. Sfenthourakis, P. Pafilis, A. Parmakelis, N. Poulakakis, and K. A. Triantis, 169–190. Broken Hill Publishers Ltd.
- Lynch, A. J., S. J. Cooke, A. H. Arthington, et al. 2023. "People Need Freshwater Biodiversity." *WIREs Water* 10, no. 3: e1633. <https://doi.org/10.1002/wat2.1633>.
- Maasri, A., S. C. Jähnig, M. C. Adamescu, et al. 2022. "A Global Agenda for Advancing Freshwater Biodiversity Research." *Ecology Letters* 25: 255–263. <https://doi.org/10.1111/ele.13931>.
- MacArthur, R. H., and E. O. Wilson. 1967. *The Theory of Island Biogeography*. Princeton Univ. Press.
- Maechler, M., P. Rousseeuw, A. Struyf, M. Hubert, and K. Hornik. 2025. "cluster: Cluster Analysis Basics and Extensions. R Package Version 2.1.8.1." <https://CRAN.R-project.org/package=cluster>.
- Marcoulides, K. M., and T. Raykov. 2019. "Evaluation of Variance Inflation Factors in Regression Models Using Latent Variable Modeling Methods." *Educational and Psychological Measurement* 79, no. 5: 874–882. <https://doi.org/10.1177/0013164418817803>.
- Meyer, P. E. 2008. "infotheo: Information-Theoretic Measures. R Package Version 1.2.0." <https://CRAN.R-project.org/package=infotheo>.
- Mouillot, D., J. De Bortoli, F. Leprieur, V. Parravicini, M. Kulbicki, and D. R. Bellwood. 2013. "The Challenge of Delineating Biogeographical Regions: Nestedness Matters for Indo-Pacific Coral Reef Fishes." *Journal of Biogeography* 40: 2228–2237. <https://doi.org/10.1111/jbi.12194>.
- Naimi, B., N. A. S. Hamm, T. A. Groen, A. K. Skidmore, and A. G. Toxopeus. 2014. "Where Is Positional Uncertainty a Problem for Species Distribution Modelling?" *Ecography* 37, no. 2: 191–203. <https://doi.org/10.1111/j.1600-0587.2013.00205.x>.
- Oikonomou, A., F. Leprieur, and I. D. Leonardos. 2014. "Biogeography of Freshwater Fishes of the Balkan Peninsula." *Hydrobiologia* 738: 205–220. <https://doi.org/10.1007/s10750-014-1930-5>.
- Oksanen, J., F. G. Blanchet, R. Kindt, et al. 2015. "vegan: Community Ecology Package. R Package Version 2.3-2." <http://CRAN.R-project.org/package=vegan>.
- Pafilis, P., I. Anastasiou, K. Sagonas, and E. Valakos. 2013. "Grazing by Goats on Islands Affects the Populations of an Endemic Mediterranean Lizard." *Journal of Zoology* 290, no. 4: 255–264. <https://doi.org/10.1111/jzo.12032>.
- Panitsa, M., A. Kagiampaki, and K. Kougioumoutzis. 2018. "Plant Diversity and Biogeography of the Aegean Archipelago: A New Synthesis." In *Biogeography and Biodiversity of the Aegean*. In Honour of Prof. Moysis Mylonas, edited by S. Sfenthourakis, P. Pafilis, A. Parmakelis, N. Poulakakis, and K. A. Triantis, 279–290. Broken Hill Publishers Ltd.
- Panitsa, M., P. Trigas, G. Iatrou, and S. Sfenthourakis. 2010. "Factors Affecting Plant Species Richness and Endemism on Land-Bridge Islands – An Example From the East Aegean Archipelago." *Acta Oecologica* 36: 431–437. <https://doi.org/10.1016/j.actao.2010.04.004>.
- Parmakelis, A., D. Dimitriadou, E. Gkigkiza, et al. 2022. "The Evolutionary History of the Relict Scorpion Family Iuridae of the Eastern Mediterranean." *Molecular Phylogenetics and Evolution* 177: 107622. <https://doi.org/10.1016/j.ympev.2022.107622>.
- Parmakelis, A., and M. Mylonas. 2004. "Dispersal and Population Structure of Two Sympatric Species of the Mediterranean Land Snail Genus *Mastus* (Gastropoda, Pulmonata, Enidae)." *Biological Journal of the Linnean Society* 83, no. 1: 131–144. <https://doi.org/10.1111/j.1095-8312.2004.00375.x>.
- Parmesan, C., M. D. Morecroft, Y. Trisurat, et al. 2022. "Terrestrial and Freshwater Ecosystems and Their Services." In *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by H. O. Pörtner, D. C. Roberts, M. Tignor, et al., 197–377. Cambridge University Press. <https://doi.org/10.1017/9781009325844.004>.
- Patiño, J., R. J. Whittaker, P. A. V. Borges, et al. 2017. "A Roadmap for Island Biology: 50 Fundamental Questions After 50 Years of the Theory of Island Biogeography." *Journal of Biogeography* 44: 963–983. <https://doi.org/10.1111/jbi.12986>.
- Patrick, C. J., M. J. Cooper, and D. G. Uzarski. 2014. "Dispersal Mode and Ability Affect the Spatial Turnover of a Wetland Macroinvertebrate Metacommunity." *Wetlands* 34, no. 6: 1133–1143. <https://doi.org/10.1007/s13157-014-0572-8>.
- Perissoratis, C., and N. Conispoliatis. 2003. "The Impacts of Sea-Level Changes During Latest Pleistocene and Holocene Times on the Morphology of the Ionian and Aegean Seas (SE Alpine Europe)." *Marine Geology* 196, no. 3–4: 145–156. [https://doi.org/10.1016/S0025-3227\(03\)00047-1](https://doi.org/10.1016/S0025-3227(03)00047-1).
- Poulakakis, N., P. Kapli, P. Lymberakis, et al. 2015. "A Review of Phylogeographic Analyses of Animal Taxa From the Aegean and Surrounding Regions." *Journal of Zoological Systematics and Evolutionary Research* 53: 18–32. <https://doi.org/10.1111/jzs.12071>.
- R Development Core Team. 2023. *R: A Language and Environment for Statistical Computing*. Version 4.3.1. R Foundation for Statistical Computing. <http://www.R-project.org/>.
- Rousseeuw, P. J. 1987. "Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis." *Journal of Computational and Applied Mathematics* 20: 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- Rueda, M., M. A. Rodriguez, and B. A. Hawkins. 2010. "Towards a Biogeographic Regionalization of the European Biota." *Journal of Biogeography* 37: 2067–2076. <https://doi.org/10.1111/j.1365-2699.2010.02388.x>.
- Salavert, V. 2013. *Life Strategies and Evolutionary Implications of Caddisflies (O. trichoptera, CL. Insecta) From Seasonal Watercourses*, 246. University of Granada. <http://hdl.handle.net/10481/24549>.
- Santi, F., R. Testolin, P. Zannini, et al. 2024. "MEDIS—A Comprehensive Spatial Database on Mediterranean Islands for Biogeographical and Evolutionary Research." *Global Ecology and Biogeography* 33, no. 8: e13855. <https://doi.org/10.1111/geb.13855>.

- Saraçlı, S., N. Doğan, and İ. Doğan. 2013. "Comparison of Hierarchical Cluster Analysis Methods by Cophenetic Correlation." *Journal of Inequalities and Applications* 2013: 203. <https://doi.org/10.1186/1029-242X-2013-203>.
- Sarremejane, R., N. Cid, R. Stubbington, et al. 2020. "DISPERSE, a Trait Database to Assess the Dispersal Potential of European Aquatic Macroinvertebrates." *Sci Data* 7: 386. <https://doi.org/10.1038/s41597-020-00732-7>.
- Sclater, P. L. 1858. "On the General Geographical Distribution of the Members of the Class Aves." *Zoological Journal of the Linnean Society* 2: 137–145. <https://doi.org/10.1111/j.1096-3642.1858.tb02549a.x>.
- Scrucca, L., C. Fraley, T. B. Murphy, and A. E. Raftery. 2023. *Model-Based Clustering, Classification, and Density Estimation Using Mclust* in R. Chapman and Hall/CRC. <https://doi.org/10.1201/9781003277965>.
- Sfenthourakis, S. 1996. "A Biogeographical Analysis of Terrestrial Isopods (Isopoda, Oniscidea) From the Central Aegean Islands (Greece)." *Journal of Biogeography* 23: 687–698. <https://doi.org/10.1111/j.1365-2699.1996.tb00029.x>.
- Sfenthourakis, S., and K. A. Triantis. 2017. "The Aegean Archipelago: A Natural Laboratory of Evolution, Ecology and Civilizations." *Journal of Biological Research-Thessaloniki* 24: 4. <https://doi.org/10.1186/s40709-017-0061-3>.
- Shiklomanov, L. A. 1993. "World Freshwater Resources." In *Water in Crisis: A Guide to World's Freshwater Resources*, edited by P. H. Gleick, 13–24. Oxford University Press.
- Sokal, R. R., and F. J. Rohlf. 1962. "The Comparison of Dendrograms by Objective Methods." *Taxon* 11, no. 2: 33–40. <https://doi.org/10.2307/1217208>.
- Soria, M., N. Cid, J. Ortega, et al. 2024. "Analysing the Contribution of Intermittent Rivers to Beta Diversity Can Improve Freshwater Conservation in Mediterranean Rivers." *Aquatic Sciences* 86: 66. <https://doi.org/10.1007/s00027-024-01074-x>.
- Strid, A. 1996. "Phytogeographia Aegaea and the Flora Hellenica Database." *Annalen Des Naturhistorischen Museums in Wien Serie B Botanik Und Zoologie* 98: 279–289.
- Tickner, D., J. J. Opperman, R. Abell, et al. 2020. "Bending the Curve of Global Freshwater Biodiversity Loss: An Emergency Recovery Plan." *Bioscience* 70, no. 4: 330–342. <https://doi.org/10.1093/biosci/biaa002>.
- Triantis, K. A., K. Kougiumoutzis, A. Legakis, et al. 2018. "The Zoogeographic Regions of the Aegean Sea: A Multi-Taxon Approach." In *Biogeography and Biodiversity of the Aegean. In Honour of Prof. Moysis Mylonas*, edited by S. Sfenthourakis, P. Pafilis, A. Parmakelis, N. Poulakakis, and K. A. Triantis, 279–290. Broken Hill Publishers Ltd.
- Triantis, K. A., and M. Mylonas. 2009. "Greek Islands, Biology." In *Encyclopedia of Islands*, edited by R. Gillespie and D. A. Glague, 388–392. University of California Press.
- Triantis, K. A., M. Mylonas, M. D. Weiser, K. Lika, and K. Vardinoyannis. 2005. "Species Richness, Environmental Heterogeneity and Area: A Case Study Based on Land Snails in Skyros Archipelago (Aegean Sea, Greece)." *Journal of Biogeography* 32: 1727–1735. <https://doi.org/10.1111/j.1365-2699.2005.01322.x>.
- Trigas, P., M. Panitsa, and S. Tsiftsis. 2013. "Elevational Gradient of Vascular Plant Species Richness and Endemism in Crete—The Effect of Post-Isolation Mountain Uplift on a Continental Island System." *PLoS One* 8: e59425. <https://doi.org/10.1371/journal.pone.0059425>.
- Vaitis, M., H. Feidas, P. Symeonidis, et al. 2019. "Development of a Spatial Database and Web-GIS for the Climate of Greece." *Earth Science Informatics* 12: 97–115. <https://doi.org/10.1007/s12145-018-0351-9>.
- Vaitis, M., V. Kopsachilis, G. Papapanos, D. Simos, N. Koukourouvli, and S. Karampela. 2017. "Supporting Island Research and Development Through the 'Archipelago' Information System." *Island Studies Journal* 12, no. 1: 95–114. <https://doi.org/10.24043/isj.7>.
- Van Andel, T. H., and J. C. Shackleton. 1982. "Late Paleolithic and Mesolithic Coastlines of Greece and the Aegean." *Journal of Field Archaeology* 9: 445–454. <https://doi.org/10.2307/529681>.
- Venables, W. N., and B. D. Ripley. 2002. *Modern Applied Statistics With S*. 4th ed. Springer.
- Vilhena, D. A., and A. Antonelli. 2015. "A Network Approach for Identifying and Delimiting Biogeographical Regions." *Nature Communications* 6: 6848. <https://doi.org/10.1038/ncomms7848>.
- Wallace, A. R. 1876. *The Geographical Distribution of Animals. With a Study of the Relations of Living and Extinct Faunas as Elucidating the Past Changes of the Earth's Surface*. Vol. 1. Harper & Brothers. <https://doi.org/10.5962/bhl.title.11354>.
- Whittaker, R. J., J. M. Fernández-Palacios, T. J. Matthews, M. K. Borregaard, and K. A. Triantis. 2017. "Island Biogeography: Taking the Long View of Nature's Laboratories." *Science* 357: eaam8326. <https://doi.org/10.1126/science.aam8326>.
- Whittaker, R. J., K. A. Triantis, and R. J. Ladle. 2008. "ORIGINAL ARTICLE: A General Dynamic Theory of Oceanic Island Biogeography." *Journal of Biogeography* 35: 977–994. <https://doi.org/10.1111/j.1365-2699.2008.01892.x>.
- Wikelski, M., D. Moskowicz, J. S. Adelman, J. Cochran, D. S. Wilcove, and M. L. May. 2006. "Simple Rules Guide Dragonfly Migration." *Biology Letters* 2, no. 3: 325–329. <https://doi.org/10.1098/rsbl.2006.0487>.
- WWF. 2022. *Living Planet Report 2022—Building a Naturepositive Society*. WWF.
- Zogaris, S., A. Economou, and P. Dimopoulos. 2009. "Ecoregions in the Southern Balkans: Should They Be Revised?" *Environmental Management* 43: 682–697. <https://doi.org/10.1007/s00267-008-9243-y>.
- Zogaris, S., and N. A. Economou. 2018. "The Biogeographic Characteristics of the River Basins of Greece." In *The Rivers of Greece: Evolution, Current Status and Perspectives*, edited by N. Skoulikidis, E. Dimitriou, I. Karaouzas, et al., vol. 59, 53–96. Hdb Env Chem. https://doi.org/10.1007/698_2017_475.
- Zuur, A. F., E. N. Ieno, and C. S. Elphick. 2010. "A Protocol for Data Exploration to Avoid Common Statistical Problems." *Methods in Ecology and Evolution* 1: 3–14. <https://doi.org/10.1111/j.2041-210X.2009.00001.x>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** jbi70117-sup-0001-Supinfo1.docx. **Data S2:** jbi70117-sup-0002-Supinfo2.docx.